

HIGH RISK ACTION

AUTHORIZATION RESULT



AUTHORIZED

HUMAN AUTHORIZER



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AUTHORIZATION RESULT



DENIED

HUMAN AUTHORIZER



AUTHORIZER UNAVAILABLE
Escalation Required

AUTHORIZATION RESULT



AUTHORIZED

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AUTHORIZATION RESULT



DENIED

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AUTHORIZER UNAVAILABLE
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EMILIA

Automated control for AI-agent actions.

How EMILIA works · No receipt, no execution.

THE SHIFT

Software stopped asking permission. It just **acts** now.

AI agents already release payments, push deploys, change permissions, curtail datacenters, and command robots — at machine speed, with no human in the loop. The actions are consequential and often **irreversible**. The question every regulator, insurer, and board now asks is the one no one can answer: **who authorized that, and can you prove it?**

THE GAP

**"A human approved this"
is a claim no one can check.**

Today that proof lives in the operator's own **editable logs** — forgeable, backfillable, and un-auditable by a regulator, court, or insurer without trusting the very party under review. Logs tell you what happened. They cannot prove **who authorized it**. That evidence gap is a category — and it is unfilled.

WHAT EMILIA IS

Automated control for every **high-risk agent action**.

Think passport control for AI agents. Before an agent may take a consequential action, EMILIA checks it: is there a valid authorization tracing to a **named human** who approved *this exact action*? If yes — **AUTHORIZED**, proceed. If not — **DENIED**. EMILIA is not the agent's identity and not the tool it calls. It is the checkpoint and the stamp.

AUTHORIZED

named human authorizer present · signed this exact action →
proceed

DENIED

authorizer unavailable / out of scope → **action does not execute**

THE ARTIFACT

A receipt that proves a named human authorized **that exact action**.

TRUST-RECEIPT · VERIFIES OFFLINE

```
action    release_payment $40,000
approver  jchen · Face ID ✓
bound     sha256:9f2c... (exact action)
sig       ed25519:R9w1...
```

✓ verified – no account, no server, no trust in us

No receipt, no execution. Change one byte → verification fails.

HOW IT WORKS

Five steps. **Deny-by-default** until proven.

- 1 **Agent requests a consequential action.** EMILIA intercepts at the actuator boundary and returns **428 — authorization required.**
- 2 **A named human signs the exact action** on their own device (Face ID / passkey). The signature is bound to a hash of the specific action, not a blanket approval.
- 3 **EMILIA verifies** the receipt: valid signature, in-scope, sufficiently assured, not replayed.
- 4 **The action executes** — only now.
- 5 **An execution receipt is emitted** — portable proof it ran, verifiable by anyone, offline.

THE PRODUCT

EMILIA Gate — the **Consequence Firewall**.

Deny-by-default at the actuator boundary. A consequential action runs only with a valid, in-scope, sufficiently-assured, non-replayed receipt — then emits proof it ran. A new security category: not antivirus (is this file bad?), not a firewall (is this packet allowed?) — **is this machine allowed to take this consequence, and can you prove who authorized it?**

Software — MCP tool calls, agent runtimes, CI/CD, systems of record.

Cloud — API-level enforcement in front of any consequential endpoint.

Physical — robots and actuators: no release without a verified receipt.

WHERE IT FITS

Complementary to all. Competitive with none.

The agent-security stack has layers. EMILIA is the one none of the others produce: proof a **named human authorized this exact action**.

IDENTITY / NHI

which machine is acting (the passport)

ACTION AUTH

which tools the agent may call (the visa)

EMILIA

which **named human** authorized **this exact action** – verifiable offline (passport control + the stamp)

Plugs into **MCP**, agent runtimes, **SCITT**, and systems of record.

WHY IT'S TRUSTWORTHY

Verifiable by anyone. Offline. Without trusting us.

A receipt is a self-contained cryptographic object. A regulator, court, insurer, or counterparty can verify it on their own machine — no EMILIA account, no server call, no trust in the operator or in us. Independent verifiers in **JavaScript**, **Python**, and **Go** agree byte-for-byte.

3

independent verifier implementations (JS
· Py · Go)

Offline

no server, no account, no phone-home

One-time

replay-defended — a receipt authorizes
one action

1 byte

change anything → verification fails

OPEN & STANDARDS-BASED

An open protocol, not a **black box**.

EMILIA is Apache-2.0 and built on open standards so it can become shared infrastructure, not a proprietary silo. The reference runtime, verifiers, and conformance vectors are public and reproducible — and the receipt composes into the emerging SCITT agent-action work (permits, action receipts, capsules) as the one leg none of them produce: named-human authorization.

Apache-2.0

open reference runtime & verifiers

IETF

authorization-receipts · quorum · SCITT
authorization-evidence profile

RR-1

public conformance suite & vectors

4,541

automated tests passing

GET STARTED

Wrap one dangerous action. See it get blocked.

TRY IT OFFLINE · 30 SECONDS

```
npx -y @emilia-protocol/crash-test
```

Spec & quickstart — emiliaprotocol.ai/spec · [/quickstart](https://emiliaprotocol.ai/quickstart)

Live demo — emiliaprotocol.ai/demo · verify a receipt at [/explorer](https://emiliaprotocol.ai/explorer)

Talk to us — team@emiliaprotocol.ai

No receipt, no execution.